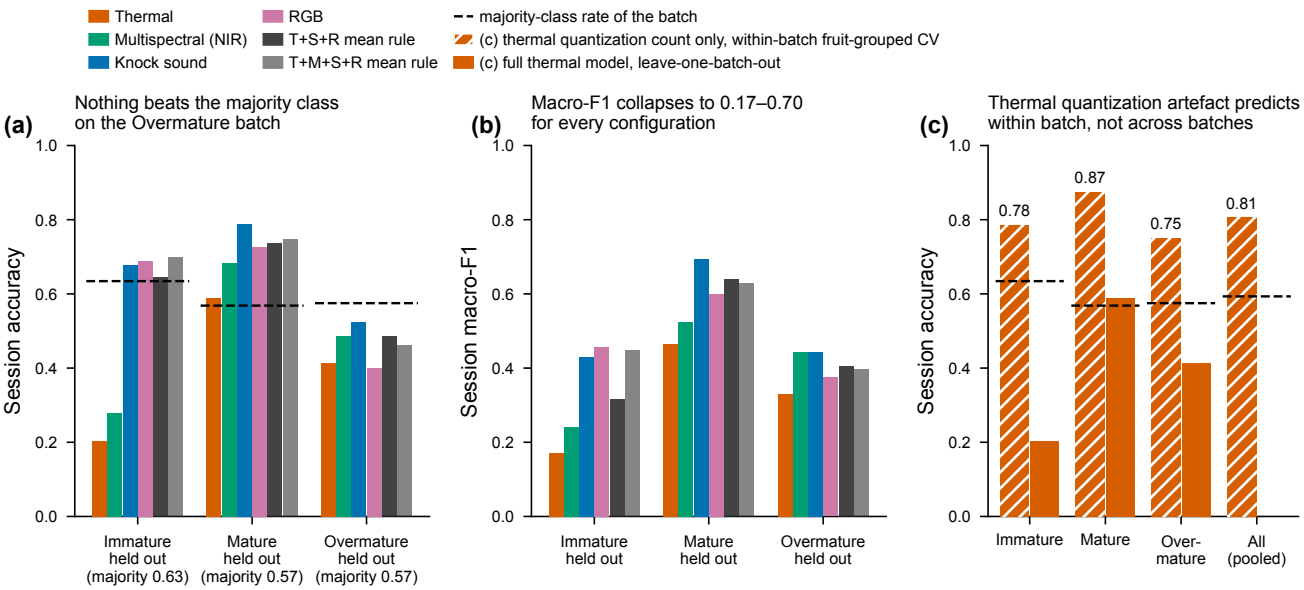
Thermal
Multispectral (NIR)
Knock sound
RGB
T+S+R mean rule
T+M+S+R mean rule
majority-class rate of the batch
(c) thermal quantization count only, within-batch fruit-grouped CV
(c) full thermal model, leave-one-batch-out
(a)
Nothing beats the majority class
on the Overmature batch
Session accuracy
1.0
0.8
0.6
0.4
0.2
0.0
Immature
held out
(majority 0.63)
Mature
held out
(majority 0.57)
Overmature
held out
(majority 0.57)
(b)
Macro-F1 collapses to 0.17–0.70
for every configuration
Session macro-F1
Immature
held out
Mature
held out
Overmature
held out
(c)
Thermal quantization artefact predicts
within batch, not across batches
Session accuracy
0.78
0.87
0.75
0.81
Immature
Mature
Over-
mature
All
(pooled)